# Peak Prediction Using Quantile Regression and Asymmetric Laplace Noise: Data-Driven Intervals for the Peak Location

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**Abstract**

Decisions in environmental and epidemiological monitoring often have to be made before a peak is fully observed. We model a series as a Gaussian trend with additive noise and estimate its amplitude, location and spread by quantile regression; for fixed location and spread the amplitude estimator is a weighted $\tau$-quantile. A $\tau$-aligned asymmetric Laplace noise model links the criterion to a likelihood and allows the signal-to-noise ratio (SNR) to be controlled. When the sequential peak-location estimate $\widehat{l}(T)$ does not stabilize early, we build candidate entry–exit intervals from the oscillations of the estimation path alone, and select one by requiring that it contain the current estimate. A Monte Carlo study ($M = 1000$) shows that the selected interval always contains the true peak at high and moderate SNR, but only about 56–84% of the time at SNR $= 1$ (horizon $T = 30$). An application to COVID-19 case counts in China, France and Germany illustrates the construction.

**Keywords:** peak prediction; quantile regression; asymmetric Laplace noise; oscillation; interval estimation; time series.

## 1 Motivation

Anticipating the timing of a peak (an epidemic wave, a pollution episode, a demand or load maximum) matters for resource allocation and preparedness. General-purpose forecasters such as ARIMA models or recurrent networks describe temporal dependence but are not designed to estimate *when* a future peak will occur, and little theory exists for peak prediction in time series (Stewart et al., 2023). We build on the framework of Dermoune et al. (2026): at a current horizon $T$, only $I(1), \ldots, I(T)$ are observed and the goal is to estimate the peak location $L$ before it is reached. Our contributions are (i) a quantile-regression criterion indexed by $\tau \in (0, 1)$ generalizing the least-absolute-deviation case $\tau = 1/2$; (ii) a probabilistic interpretation through $\tau$-aligned asymmetric Laplace noise; and (iii) a data-driven interval for $L$ built only from the estimation trajectory, together with an empirical assessment of its coverage.

## 2 Model and quantile-based estimation

We assume

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t), \qquad t = 1, \ldots, T, \tag{1}$$

with amplitude $A > 0$, peak location $L$, spread $S > 0$ and noise $e(t)$. With the check function $\rho_\tau(u) = u\,(\tau - \mathbf{1}_{\{u<0\}})$ we minimize

$$f_\tau(T, a, l, s) = \frac{1}{T}\sum_{t=1}^{T} \rho_\tau\Big(I(t) - a\, e^{-(t-l)^2/s^2}\Big). \tag{2}$$

For fixed $(l, s)$, put $w(t) = e^{-(t-l)^2/s^2}$ and $x(t) = I(t)/w(t)$. Since $\rho_\tau(I - aw) = w\,\rho_\tau(x - a)$, the minimizing amplitude is the *weighted $\tau$-quantile* of $x(t)$ with weights $w(t)$. The scale $s$ is then optimized numerically and $\widehat{l}(T)$ is the minimizer over $l \in \{1, \dots, h\}$, where $h > T$ is a fixed horizon; thus $\widehat{l}(T)$ may point beyond the observed window.

**Noise model.** To study the role of $\tau$ and of the SNR, we take $e(t)$ with density $f_{\tau,b}(x) = \frac{\tau(1-\tau)}{b}\exp\{-\rho_\tau(x)/b\}$. Then $\mathbb{P}(e < 0) = \tau$, the negative log-likelihood is proportional to the quantile loss, and $\tau = 1/2$ gives the symmetric Laplace law. Defining $\mathrm{SNR} = A/\sqrt{\mathrm{Var}(e)}$, the scale is calibrated as $b = \frac{A}{\mathrm{SNR}}\,\tau(1-\tau)/\sqrt{\tau^2 + (1-\tau)^2}$.

## 3 Candidate entry–exit intervals

For a current horizon $T$, define the oscillation of the path over $[t, T]$,

$$\omega_l(t, T) = \max_{t \le t_1, t_2 \le T} \big|\widehat{l}(t_1) - \widehat{l}(t_2)\big|,$$

which is nonincreasing in $t$ and vanishes at $t = T$. The *candidate exit times* are $T^{(0)}_{\mathrm{exit}}(T) = T$ followed by the strict-decrease points $t < T$ of $t \mapsto \omega_l(t, T)$ in decreasing order, $T^{(1)}_{\mathrm{exit}}(T) > T^{(2)}_{\mathrm{exit}}(T) > \cdots$, with levels $\epsilon'_k(T) = \omega_l(T^{(k)}_{\mathrm{exit}}, T)$. For each $k$, let

$$E_k(T) = \{t < T^{(k)}_{\mathrm{exit}} :\ \omega_l(t, T^{(k)}_{\mathrm{exit}}) \ge \epsilon'_k(T)\}, \qquad T^*_k = \max E_k(T), \qquad \epsilon_k = \omega_l(T^*_k, T^{(k)}_{\mathrm{exit}}),$$

and define the entry time as the smallest $t < T^{(k)}_{\mathrm{exit}}$ with $\omega_l(t, T^{(k)}_{\mathrm{exit}}) = \epsilon_k$. The candidate interval and its width are

$$\mathcal{R}_k(T) = \big[T^{(k)}_{\mathrm{entry}}(T),\, T^{(k)}_{\mathrm{exit}}(T)\big], \qquad w_k(T) = T^{(k)}_{\mathrm{exit}}(T) - T^{(k)}_{\mathrm{entry}}(T).$$

Only the trajectory $\widehat{l}(1), \dots, \widehat{l}(T)$ is used; $L$ is never required. In simulations at SNR = 1000 we observe that both endpoints are nonincreasing in $k$ (no violation in the simulated pairs), but consecutive candidates are not in general nested: they may overlap or be disjoint, and $T^{(k)}_{\mathrm{entry}} = T^{(k+1)}_{\mathrm{exit}}$ holds only for some pairs.

**Operational selection.** Let $\mathcal{K}(T) = \{k :\ T^{(k)}_{\mathrm{entry}}(T) \le \widehat{l}(T) \le T^{(k)}_{\mathrm{exit}}(T)\}$. If $\mathcal{K}(T) \neq \varnothing$ we select the narrowest admissible candidate,

$$\widehat{k}(T) = \operatorname*{arg\,min}_{k \in \mathcal{K}(T)} w_k(T), \qquad \widehat{\mathcal{I}}(T) = \mathcal{R}_{\widehat{k}(T)}(T),$$

with ties broken by smaller $k$, then smaller entry and exit times; otherwise no interval is produced at $T$. The first selection horizon is $T_{\mathrm{est}} = \min\{T : \mathcal{K}(T) \neq \varnothing\}$. Neither $\widehat{\mathcal{I}}(T)$ nor $T_{\mathrm{est}}$ uses $L$. The rule guarantees consistency with $\widehat{l}(T)$, not containment of $L$, so $\widehat{\mathcal{I}}(T)$ is a data-driven interval estimate and not a confidence interval; no coverage guarantee $\mathbb{P}\{L \in \widehat{\mathcal{I}}(T)\} \ge 1 - \alpha$ is established. Coverage is therefore assessed by simulation.

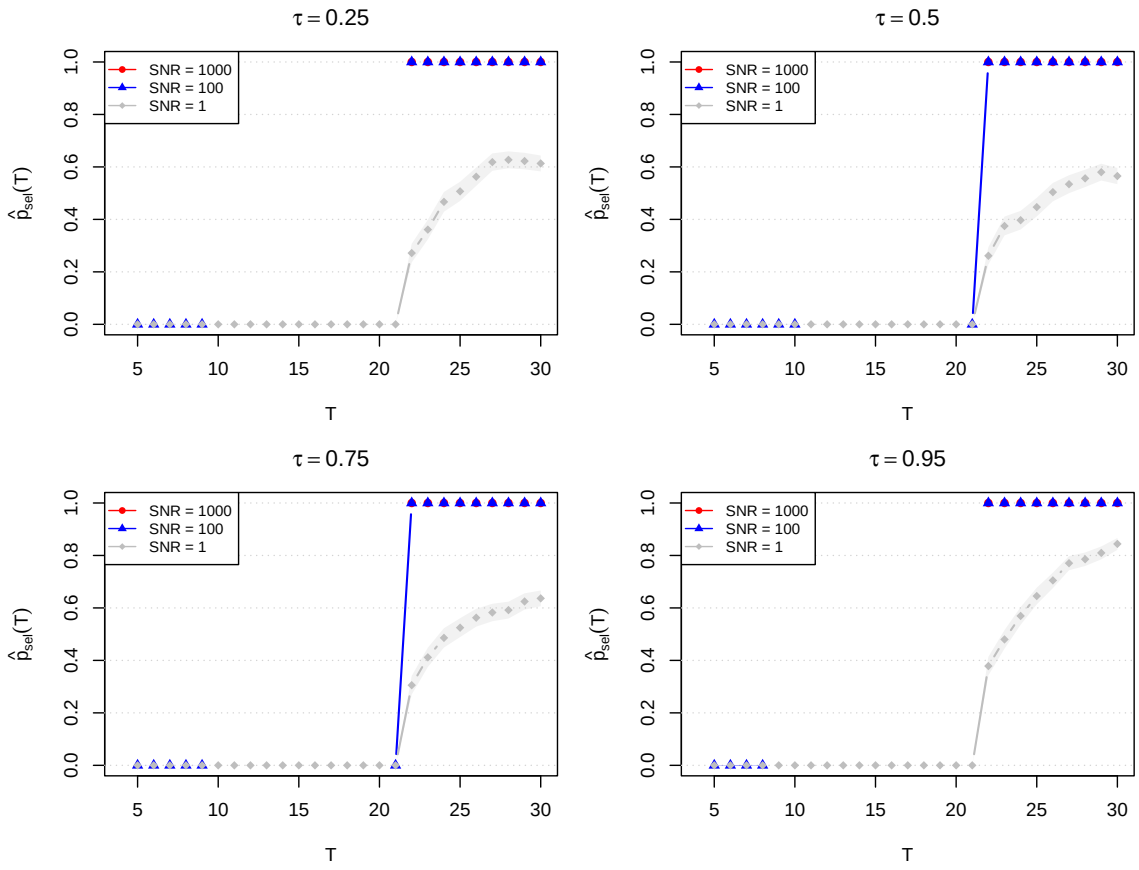



Figure 1: Empirical coverage $\widehat{p}_M^{\mathrm{sel}}(T)$ of the selected interval versus the horizon $T$ ($M = 1000$), one panel per $\tau$ and one curve per SNR; the true peak is $L = 22$.

## 4 Simulation study of empirical coverage

We take $A = 1$, $L = 22$, $S = 10$, $h = 30$, $\tau \in \{0.25, 0.50, 0.75, 0.95\}$ and SNR $\in \{1000, 100, 1\}$, with $M = 1000$ independent trajectories per setting. The selection rule is applied separately to each trajectory at each $T$; $L$ is used only afterwards, to score coverage. We report

$$\widehat{q}_M(T) = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{\widehat{\mathcal{I}}_m(T) \neq \varnothing\}, \qquad \widehat{p}_M^{\mathrm{sel}}(T) = \frac{\sum_m \mathbf{1}\{L \in \widehat{\mathcal{I}}_m(T)\}}{\sum_m \mathbf{1}\{\widehat{\mathcal{I}}_m(T) \neq \varnothing\}},$$

the selection rate and the coverage among selected intervals, together with the unconditional proportion $\widehat{p}_M^{\mathrm{all}}(T)$, in which failures to select count as misses.

The results depend strongly on the SNR (Figure 1, Table 1). At SNR $= 1000$ no interval is selected for $T < 22$, and from $T = 22$ onward both the selection rate and the coverage equal one. At SNR $= 100$, selection is rare at some early horizons, but from $T = 23$ onward every trajectory yields an interval that contains $L$. At SNR $= 1$, intervals may be selected before $T = 22$, but in our simulations none of them contains $L$; at $T = 22$ the coverage among selected intervals is only about 0.26–0.38, and it rises with $T$ to 0.57–0.84 at $T = 30$, depending on $\tau$ (best for $\tau = 0.95$). Since the selection rate at $T = 30$ is at least 0.996, the low coverage at SNR $= 1$ is not explained by failures to select. Note also that the perfect coverage at high and moderate SNR comes with wide intervals (mean width 12–20 on a horizon of 30), whereas the low-SNR intervals are narrow (about 5–6) but often miss $L$. These are empirical coverage rates for this design, not a theoretical guarantee.

Table 1: Selection rate, empirical coverage and mean width at $T = 30$ ($M = 1000$).

| $\tau$ | SNR | $\widehat{q}_M(30)$ | $\widehat{p}_M^{\mathrm{sel}}(30)$ | $\widehat{p}_M^{\mathrm{all}}(30)$ | Mean width |
|---|---|---|---|---|---|
| 0.25 | 1000 | 1.000 | 1.000 | 1.000 | 18.70 |
| 0.25 | 100 | 1.000 | 1.000 | 1.000 | 12.66 |
| 0.25 | 1 | 0.998 | 0.613 | 0.612 | 5.62 |
| 0.50 | 1000 | 1.000 | 1.000 | 1.000 | 18.26 |
| 0.50 | 100 | 1.000 | 1.000 | 1.000 | 11.98 |
| 0.50 | 1 | 0.996 | 0.565 | 0.563 | 5.49 |
| 0.75 | 1000 | 1.000 | 1.000 | 1.000 | 18.79 |
| 0.75 | 100 | 1.000 | 1.000 | 1.000 | 12.72 |
| 0.75 | 1 | 0.996 | 0.637 | 0.634 | 5.30 |
| 0.95 | 1000 | 1.000 | 1.000 | 1.000 | 19.66 |
| 0.95 | 100 | 1.000 | 1.000 | 1.000 | 14.20 |
| 0.95 | 1 | 1.000 | 0.844 | 0.844 | 5.75 |

Table 2: Retrospectively selected windows $(T_{\mathrm{entry}}, T_{\mathrm{exit}})$ strictly containing the observed peak $L$; "–" means no candidate contains $L$.

| Country | $L$ | $\tau = 0.25$ | $\tau = 0.50$ | $\tau = 0.75$ | $\tau = 0.95$ |
|---|---|---|---|---|---|
| China | 22 | (15,26) | (15,24) | – | (13,26) |
| France | 78 | (72,79) | (73,82) | (70,79) | (70,80) |
| Germany | 61 | (60,62) | – | – | (56,63) |

## 5 Application to COVID-19 case counts

We apply the procedure to daily confirmed cases in China (22 January–22 March 2020; JHU CSSE), France and Germany (ECDC), for $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. Because the observed peak $L$ is known retrospectively in each series, the candidates can be evaluated ex post: among those with $T_{\mathrm{entry}}^{(k)} < L < T_{\mathrm{exit}}^{(k)}$ we report the narrowest (this uses $L$ and is not part of the operational rule). Table 2 gives the resulting windows.

For France, all four quantile levels give a window containing the peak, with widths between 7 and 10. For China the windows are $(15, 24)$ at $\tau = 0.5$, close to the window $(14, 25)$ obtained by Dermoune et al. (2026) under the least-absolute-deviation criterion, while $\tau = 0.75$ yields no containing candidate. For Germany only $\tau = 0.25$ and $\tau = 0.95$ do. The window therefore depends on the quantile level, and a containing window is not always available.

## 6 Conclusion and perspectives

Quantile regression with $\tau$-aligned asymmetric Laplace noise gives a unified criterion for peak-location estimation, and the oscillation-based entry–exit construction yields a data-driven interval for $L$ that is computed without knowledge of $L$. Simulation shows that its coverage is excellent but costly in width at high and moderate SNR, and limited at low SNR. Open questions are a theoretical coverage result (or a calibration of the selection rule to reach a target level), repeating the study across other peak locations and widths, and applying it to environmental series such as pollution or energy-demand peaks.

## References

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